

Creation Date 11-Feb-2010

Revision Date 10-Feb-2024

Revision Number 4

## SECTION 1: IDENTIFICATION OF THE SUBSTANCE/MIXTURE AND OF THE COMPANY/UNDERTAKING

### 1.1. Product identifier

|                           |                        |
|---------------------------|------------------------|
| Product Description:      | <b>Nickel Thinfoil</b> |
| Cat No. :                 | <b>35814</b>           |
| Index No                  | 028-002-00-7           |
| CAS No                    | 7440-02-0              |
| Molecular Formula         | Ni                     |
| REACH registration number | -                      |

### 1.2. Relevant identified uses of the substance or mixture and uses advised against

|                      |                          |
|----------------------|--------------------------|
| Recommended Use      | Laboratory chemicals.    |
| Uses advised against | No Information available |

### 1.3. Details of the supplier of the safety data sheet

#### Company

Thermo Fisher (Kandel) GmbH  
Erlenbachweg 2, 76870 Kandel, Germany  
Tel: +49 (0) 721 84007 280  
Fax: +49 (0) 721 84007 300

**Swiss distributor** - Fisher Scientific AG  
Neuhofstrasse 11, CH 4153 Reinach  
Tel: +41 (0) 56 618 41 11  
<https://www.fishersci.ch/ch/en/customer-help-support/forms/email-us.html>

**E-mail address** begel.sdsdesk@thermofisher.com

### 1.4. Emergency telephone number

For information **US** call: 001-800-227-6701 / **Europe** call: +32 14 57 52 11  
Emergency Number **US**:001-201-796-7100 / **Europe**: +32 14 57 52 99  
**CHEMTREC** Tel. No.**US**:001-800-424-9300 / **Europe**:001-703-527-3887

customers in Switzerland:  
Tox Info Suisse Emergency Number: **145 (24hr)**  
Tox Info Suisse: +41-44 251 51 51 (Emergency number from abroad)  
Chemtrec (24h) Toll-Free: 0800 564 402  
Chemtrec Local: +41-43 508 20 11 (Zurich)

## SECTION 2: HAZARDS IDENTIFICATION

### 2.1. Classification of the substance or mixture

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## CLP Classification - Regulation (EC) No 1272/2008

### Physical hazards

Based on available data, the classification criteria are not met

### Health hazards

Skin Sensitization

Category 1 (H317)

Carcinogenicity

Category 2 (H351)

Specific target organ toxicity - (repeated exposure)

Category 1 (H372)

### Environmental hazards

Based on available data, the classification criteria are not met

Full text of Hazard Statements: see section 16

## 2.2. Label elements



Signal Word

**Danger**

### **Hazard Statements**

H317 - May cause an allergic skin reaction

H351 - Suspected of causing cancer

H372 - Causes damage to organs through prolonged or repeated exposure

### **Precautionary Statements**

P201 - Obtain special instructions before use

P280 - Wear protective gloves/protective clothing/eye protection/face protection

P302 + P352 - IF ON SKIN: Wash with plenty of soap and water

P308 + P313 - IF exposed or concerned: Get medical advice/attention

P362 + P364 - Take off contaminated clothing and wash it before reuse

## 2.3. Other hazards

In accordance with Annex XIII of the REACH Regulation, inorganic substances do not require assessment

Toxicity to Soil Dwelling Organisms

This product does not contain any known or suspected endocrine disruptors

## **SECTION 3: COMPOSITION/INFORMATION ON INGREDIENTS**

### 3.1. Substances

| Component | CAS No    | EC No             | Weight % | CLP Classification - Regulation (EC) No 1272/2008 |
|-----------|-----------|-------------------|----------|---------------------------------------------------|
| Nickel    | 7440-02-0 | EEC No. 231-111-4 | >95      | Skin Sens. 1 (H317)<br>Carc. 2 (H351)             |

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|  |  |  |  |                  |
|--|--|--|--|------------------|
|  |  |  |  | STOT RE 1 (H372) |
|--|--|--|--|------------------|

|                           |   |
|---------------------------|---|
| REACH registration number | - |
|---------------------------|---|

Full text of Hazard Statements: see section 16

## SECTION 4: FIRST AID MEASURES

### 4.1. Description of first aid measures

|                                    |                                                                                                                                                                                                                                                                                                                                |
|------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| General Advice                     | Show this safety data sheet to the doctor in attendance. Immediate medical attention is required.                                                                                                                                                                                                                              |
| Eye Contact                        | Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes. In the case of contact with eyes, rinse immediately with plenty of water and seek medical advice.                                                                                                                                     |
| Skin Contact                       | Wash off immediately with plenty of water for at least 15 minutes. Immediate medical attention is required.                                                                                                                                                                                                                    |
| Ingestion                          | Do NOT induce vomiting. Call a physician or poison control center immediately.                                                                                                                                                                                                                                                 |
| Inhalation                         | Remove to fresh air. If not breathing, give artificial respiration. Do not use mouth-to-mouth method if victim ingested or inhaled the substance; give artificial respiration with the aid of a pocket mask equipped with a one-way valve or other proper respiratory medical device. Immediate medical attention is required. |
| Self-Protection of the First Aider | Ensure that medical personnel are aware of the material(s) involved, take precautions to protect themselves and prevent spread of contamination.                                                                                                                                                                               |

### 4.2. Most important symptoms and effects, both acute and delayed

May cause allergic skin reaction. Symptoms of allergic reaction may include rash, itching, swelling, trouble breathing, tingling of the hands and feet, dizziness, lightheadedness, chest pain, muscle pain or flushing

### 4.3. Indication of any immediate medical attention and special treatment needed

|                    |                        |
|--------------------|------------------------|
| Notes to Physician | Treat symptomatically. |
|--------------------|------------------------|

## SECTION 5: FIREFIGHTING MEASURES

### 5.1. Extinguishing media

#### Suitable Extinguishing Media

Use extinguishing measures that are appropriate to local circumstances and the surrounding environment. Water spray, carbon dioxide (CO<sub>2</sub>), dry chemical, alcohol-resistant foam.

#### Extinguishing media which must not be used for safety reasons

No information available.

### 5.2. Special hazards arising from the substance or mixture

Thermal decomposition can lead to release of irritating gases and vapors.

#### Hazardous Combustion Products

None under normal use conditions.

### 5.3. Advice for firefighters

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As in any fire, wear self-contained breathing apparatus pressure-demand, MSHA/NIOSH (approved or equivalent) and full protective gear. Thermal decomposition can lead to release of irritating gases and vapors.

## SECTION 6: ACCIDENTAL RELEASE MEASURES

### 6.1. Personal precautions, protective equipment and emergency procedures

Use personal protective equipment as required. Ensure adequate ventilation. Avoid dust formation. Keep people away from and upwind of spill/leak. Evacuate personnel to safe areas.

### 6.2. Environmental precautions

Do not flush into surface water or sanitary sewer system.

### 6.3. Methods and material for containment and cleaning up

Sweep up and shovel into suitable containers for disposal. Avoid dust formation.

### 6.4. Reference to other sections

Refer to protective measures listed in Sections 8 and 13.

## SECTION 7: HANDLING AND STORAGE

### 7.1. Precautions for safe handling

Wear personal protective equipment/face protection. Avoid dust formation. Do not get in eyes, on skin, or on clothing. Use only under a chemical fume hood. Do not breathe (dust, vapor, mist, gas). Do not ingest. If swallowed then seek immediate medical assistance.

#### **Hygiene Measures**

Handle in accordance with good industrial hygiene and safety practice. Keep away from food, drink and animal feeding stuffs. Do not eat, drink or smoke when using this product. Remove and wash contaminated clothing and gloves, including the inside, before re-use. Wash hands before breaks and after work.

### 7.2. Conditions for safe storage, including any incompatibilities

Keep containers tightly closed in a dry, cool and well-ventilated place.

**Technical Rules for Hazardous Substances (TRGS) 510**  
**Storage Class (LGK) (Germany)**

Storage Class/LGK 6.1D

**Switzerland - Storage of hazardous substances**

Storage class - SC 6.1  
<https://www.kvu.ch/de/themen/stoffe-und-produkte>  
<https://www.kvu.ch/fr/themes/substances-et-produits>  
<https://www.kvu.ch/it/temi/sostanze-e-prodotti>

### 7.3. Specific end use(s)

Use in laboratories

## SECTION 8: EXPOSURE CONTROLS/PERSONAL PROTECTION

### 8.1. Control parameters

#### **Exposure limits**

List source(s): **UK** - EH40/2005 Work Exposure Limits, Forth edition. Published 2020. **IRE** - 2010 Code of Practice for the Safety, Health and Welfare at Work (Chemical Agents) Regulations 2001. Published by the Health and Safety Authority. **CH** - The Government of Switzerland has set a directive on limit values for working materials (Grenzwerte am Arbeitsplatz) which is

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based on the Swiss Federal Regulation "Verordnung über die Verhütung von Unfällen und Berufskrankheiten". This directive is administered, periodically revised and enforced by SUVA (Swiss National Accident Insurance Fund).

| Component | European Union | The United Kingdom                                                            | France                                                                                                  | Belgium                         | Spain                                       |
|-----------|----------------|-------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------|---------------------------------|---------------------------------------------|
| Nickel    |                | STEL: 1.5 mg/m <sup>3</sup> 15 min<br>TWA: 0.5 mg/m <sup>3</sup> 8 hr<br>Skin | TWA / VME: 1 mg/m <sup>3</sup> (8 heures).<br>TWA / VME: 1 mg/m <sup>3</sup> (8 heures). metal gratings | TWA: 1 mg/m <sup>3</sup> 8 uren | TWA / VLA-ED: 1 mg/m <sup>3</sup> (8 horas) |

| Component | Italy | Germany                                                                                                                               | Portugal                           | The Netherlands | Finland                                |
|-----------|-------|---------------------------------------------------------------------------------------------------------------------------------------|------------------------------------|-----------------|----------------------------------------|
| Nickel    |       | TWA: 0.03 mg/m <sup>3</sup> (8 Stunden). AGW - exposure factor 8<br>TWA: 0.006 mg/m <sup>3</sup> (8 Stunden). AGW - exposure factor 8 | TWA: 1.5 mg/m <sup>3</sup> 8 horas |                 | TWA: 0.01 mg/m <sup>3</sup> 8 tunteina |

| Component | Austria                                                                    | Denmark                                                                        | Switzerland                          | Poland                                  | Norway                                                                                            |
|-----------|----------------------------------------------------------------------------|--------------------------------------------------------------------------------|--------------------------------------|-----------------------------------------|---------------------------------------------------------------------------------------------------|
| Nickel    | TRK-KZGW: 2 mg/m <sup>3</sup> 15 Minuten<br>TRK-TMW: 0.5 mg/m <sup>3</sup> | TWA: 0.05 mg/m <sup>3</sup> 8 timer<br>STEL: 0.1 mg/m <sup>3</sup> 15 minutter | TWA: 0.5 mg/m <sup>3</sup> 8 Stunden | TWA: 0.25 mg/m <sup>3</sup> 8 godzinach | TWA: 0.05 mg/m <sup>3</sup> 8 timer<br>STEL: 0.15 mg/m <sup>3</sup> 15 minutter. value calculated |

| Component | Bulgaria                    | Croatia                                  | Ireland                                                                | Cyprus | Czech Republic                                                                                        |
|-----------|-----------------------------|------------------------------------------|------------------------------------------------------------------------|--------|-------------------------------------------------------------------------------------------------------|
| Nickel    | TWA: 0.05 mg/m <sup>3</sup> | TWA-GVI: 0.5 mg/m <sup>3</sup> 8 satima. | TWA: 0.5 mg/m <sup>3</sup> 8 hr.<br>STEL: 1.5 mg/m <sup>3</sup> 15 min |        | TWA: 0.5 mg/m <sup>3</sup> 8 hodinách. respirable fraction of aerosol<br>Ceiling: 1 mg/m <sup>3</sup> |

| Component | Estonia                                | Gibraltar | Greece                   | Hungary                                  | Iceland                                                                                                              |
|-----------|----------------------------------------|-----------|--------------------------|------------------------------------------|----------------------------------------------------------------------------------------------------------------------|
| Nickel    | TWA: 0.5 mg/m <sup>3</sup> 8 tundides. |           | TWA: 1 mg/m <sup>3</sup> | TWA: 0.01 mg/m <sup>3</sup> 8 órában. AK | TWA: 0.05 mg/m <sup>3</sup> 8 klukkustundum. Ni dust and powder<br>Ceiling: 0.1 mg/m <sup>3</sup> Ni dust and powder |

| Component | Latvia                      | Lithuania                       | Luxembourg | Malta | Romania                                                                   |
|-----------|-----------------------------|---------------------------------|------------|-------|---------------------------------------------------------------------------|
| Nickel    | TWA: 0.05 mg/m <sup>3</sup> | TWA: 0.5 mg/m <sup>3</sup> IPRD |            |       | TWA: 0.1 mg/m <sup>3</sup> 8 ore<br>STEL: 0.5 mg/m <sup>3</sup> 15 minute |

| Component | Russia                      | Slovak Republic                                                                   | Slovenia                                                                                                                | Sweden                                   | Turkey |
|-----------|-----------------------------|-----------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------|------------------------------------------|--------|
| Nickel    | MAC: 0.05 mg/m <sup>3</sup> | TWA: 0.5 mg/m <sup>3</sup> 8 hodinách<br>STEL: 0.05 mg/m <sup>3</sup> 15 minútach | TWA: 0.006 mg/m <sup>3</sup> 8 urah respirable fraction<br>STEL: 0.048 mg/m <sup>3</sup> 15 minutah respirable fraction | TLV: 0.5 mg/m <sup>3</sup> 8 timmar. NGV |        |

## Biological limit values

List source(s):

| Component | Italy | Finland                                                                           | Denmark | Bulgaria                                        | Romania                           |
|-----------|-------|-----------------------------------------------------------------------------------|---------|-------------------------------------------------|-----------------------------------|
| Nickel    |       | Nickel: 0.1 µmol/L urine after the shift after a working week or exposure period. |         | Nickel: 45 µg/L urine after several work shifts | Nickel: 3 µg/L urine end of shift |

| Component | Gibraltar | Latvia               | Slovak Republic                                       | Luxembourg | Turkey |
|-----------|-----------|----------------------|-------------------------------------------------------|------------|--------|
| Nickel    |           | Nickel: 3 µg/L urine | Nickel: 0.03 mg/L blood end of exposure or work shift |            |        |

## Monitoring methods

BS EN 14042:2003 Title Identifier: Workplace atmospheres. Guide for the application and use of procedures for the assessment of exposure to chemical and biological agents.

MDHS14/3 General methods for sampling and gravimetric analysis of respirable and inhalable dust

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## Derived No Effect Level (DNEL) / Derived Minimum Effect Level (DMEL)

See table for values

| Component                 | Acute effects local (Dermal) | Acute effects systemic (Dermal) | Chronic effects local (Dermal) | Chronic effects systemic (Dermal) |
|---------------------------|------------------------------|---------------------------------|--------------------------------|-----------------------------------|
| Nickel<br>7440-02-0 (>95) |                              |                                 | DNEL = 0.035mg/cm2             |                                   |

| Component                 | Acute effects local (Inhalation) | Acute effects systemic (Inhalation) | Chronic effects local (Inhalation) | Chronic effects systemic (Inhalation) |
|---------------------------|----------------------------------|-------------------------------------|------------------------------------|---------------------------------------|
| Nickel<br>7440-02-0 (>95) | DNEL = 11.9mg/m <sup>3</sup>     |                                     | DNEL = 0.05mg/m <sup>3</sup>       | DNEL = 0.05mg/m <sup>3</sup>          |

## Predicted No Effect Concentration (PNEC)

See values below.

| Component                 | Fresh water    | Fresh water sediment        | Water Intermittent | Microorganisms in sewage treatment | Soil (Agriculture)       |
|---------------------------|----------------|-----------------------------|--------------------|------------------------------------|--------------------------|
| Nickel<br>7440-02-0 (>95) | PNEC = 7.1µg/L | PNEC = 109mg/kg sediment dw |                    | PNEC = 0.33mg/L                    | PNEC = 29.9mg/kg soil dw |

| Component                 | Marine water   | Marine water sediment       | Marine water Intermittent | Food chain            | Air |
|---------------------------|----------------|-----------------------------|---------------------------|-----------------------|-----|
| Nickel<br>7440-02-0 (>95) | PNEC = 8.6µg/L | PNEC = 109mg/kg sediment dw |                           | PNEC = 0.12mg/kg food |     |

## 8.2. Exposure controls

### Engineering Measures

Use only under a chemical fume hood. Ensure that eyewash stations and safety showers are close to the workstation location. Wherever possible, engineering control measures such as the isolation or enclosure of the process, the introduction of process or equipment changes to minimise release or contact, and the use of properly designed ventilation systems, should be adopted to control hazardous materials at source

### Personal protective equipment

#### Eye Protection

Wear safety glasses with side shields (or goggles) (European standard - EN 166)

#### Hand Protection

Protective gloves

| Glove material | Breakthrough time                 | Glove thickness | EU standard | Glove comments        |
|----------------|-----------------------------------|-----------------|-------------|-----------------------|
| Natural rubber | See manufacturers recommendations | -               | EN 374      | (minimum requirement) |
| Nitrile rubber |                                   |                 |             |                       |
| Neoprene       |                                   |                 |             |                       |
| PVC            |                                   |                 |             |                       |

#### Skin and body protection

Wear appropriate protective gloves and clothing to prevent skin exposure.

Inspect gloves before use. observe the instructions regarding permeability and breakthrough time which are provided by the supplier of the gloves. (Refer to manufacturer/supplier for information) gloves are suitable for the task: Chemical compatability, Dexterity, Operational conditions, User susceptibility, e.g. sensitisation effects, also take into consideration the specific local conditions under which the product is used, such as the danger of cuts, abrasion. gloves with care avoiding skin contamination.

#### Respiratory Protection

When workers are facing concentrations above the exposure limit they must use appropriate certified respirators.  
To protect the wearer, respiratory protective equipment must be the correct fit and be used and maintained properly

### Large scale/emergency use

Use a NIOSH/MSHA or European Standard EN 136 approved respirator if exposure limits are exceeded or if irritation or other symptoms are experienced

**Recommended Filter type:** Particulates filter conforming to EN 143

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**Small scale/Laboratory use** Use a NIOSH/MSHA or European Standard EN 149:2001 approved respirator if exposure limits are exceeded or if irritation or other symptoms are experienced.  
**Recommended half mask:-** Particle filtering: EN149:2001  
When RPE is used a face piece Fit Test should be conducted

**Environmental exposure controls** Prevent product from entering drains. Do not allow material to contaminate ground water system. Local authorities should be advised if significant spillages cannot be contained.

## SECTION 9: PHYSICAL AND CHEMICAL PROPERTIES

### 9.1. Information on basic physical and chemical properties

|                                                |                                                |                                          |
|------------------------------------------------|------------------------------------------------|------------------------------------------|
| <b>Physical State</b>                          | Solid                                          |                                          |
| <b>Appearance</b>                              |                                                |                                          |
| <b>Odor</b>                                    | No information available                       |                                          |
| <b>Odor Threshold</b>                          | No data available                              |                                          |
| <b>Melting Point/Range</b>                     | 1455 °C / 2651 °F                              |                                          |
| <b>Softening Point</b>                         | No data available                              |                                          |
| <b>Boiling Point/Range</b>                     | 2732 - 2837 °C / 4949.6 - 5138.6 @ 760 mmHg °F |                                          |
| <b>Flammability (liquid)</b>                   | Not applicable                                 | Solid                                    |
| <b>Flammability (solid,gas)</b>                | No information available                       |                                          |
| <b>Explosion Limits</b>                        | No data available                              |                                          |
| <b>Flash Point</b>                             | No information available                       | <b>Method -</b> No information available |
| <b>Autoignition Temperature</b>                | No data available                              |                                          |
| <b>Decomposition Temperature</b>               | No data available                              |                                          |
| <b>pH</b>                                      | No information available                       |                                          |
| <b>Viscosity</b>                               | Not applicable                                 | Solid                                    |
| <b>Water Solubility</b>                        | Insoluble                                      |                                          |
| <b>Solubility in other solvents</b>            | No information available                       |                                          |
| <b>Partition Coefficient (n-octanol/water)</b> |                                                |                                          |
| <b>Vapor Pressure</b>                          | 1 mmHg @ 1810 °C                               |                                          |
| <b>Density / Specific Gravity</b>              | No data available                              |                                          |
| <b>Bulk Density</b>                            | No data available                              |                                          |
| <b>Vapor Density</b>                           | Not applicable                                 | Solid                                    |
| <b>Particle characteristics</b>                | No data available                              |                                          |

### 9.2. Other information

|                          |                        |
|--------------------------|------------------------|
| <b>Molecular Formula</b> | Ni                     |
| <b>Molecular Weight</b>  | 58.7                   |
| <b>Evaporation Rate</b>  | Not applicable - Solid |

## SECTION 10: STABILITY AND REACTIVITY

**10.1. Reactivity** None known, based on information available

**10.2. Chemical stability** Stable under normal conditions.

### 10.3. Possibility of hazardous reactions

|                                 |                                          |
|---------------------------------|------------------------------------------|
| <b>Hazardous Polymerization</b> | Hazardous polymerization does not occur. |
| <b>Hazardous Reactions</b>      | None under normal processing.            |

### 10.4. Conditions to avoid

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Incompatible products. Excess heat. Avoid dust formation.

## 10.5. Incompatible materials

Strong oxidizing agents.

## 10.6. Hazardous decomposition products

None under normal use conditions.

## SECTION 11: TOXICOLOGICAL INFORMATION

### 11.1. Information on hazard classes as defined in Regulation (EC) No 1272/2008

#### Product Information

##### (a) acute toxicity;

Oral

Based on available data, the classification criteria are not met

Dermal

No data available

Inhalation

No data available

| Component | LD50 Oral                 | LD50 Dermal | LC50 Inhalation              |
|-----------|---------------------------|-------------|------------------------------|
| Nickel    | LD50 > 9000 mg/kg ( Rat ) | -           | LC50 > 10.2 mg/L ( Rat ) 1 h |

##### (b) skin corrosion/irritation;

No data available

##### (c) serious eye damage/irritation;

No data available

##### (d) respiratory or skin sensitization;

Respiratory

No data available

Skin

Category 1

No information available

##### (e) germ cell mutagenicity;

No data available

##### (f) carcinogenicity;

Category 2

The table below indicates whether each agency has listed any ingredient as a carcinogen

| Component | EU | UK | Germany | IARC     |
|-----------|----|----|---------|----------|
| Nickel    |    |    | Cat. 1  | Group 2B |

##### (g) reproductive toxicity;

No data available

##### (h) STOT-single exposure;

No data available

##### (i) STOT-repeated exposure;

Category 1

Target Organs

Kidney, Skin, Lungs, Blood.

##### (j) aspiration hazard;

Not applicable

Solid

Symptoms / effects, both acute and delayed

Symptoms of allergic reaction may include rash, itching, swelling, trouble breathing, tingling of the hands and feet, dizziness, lightheadedness, chest pain, muscle pain or flushing.

### 11.2. Information on other hazards



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**Endocrine Disrupting Properties** Assess endocrine disrupting properties for human health. This product does not contain any known or suspected endocrine disruptors.

## SECTION 12: ECOLOGICAL INFORMATION

### 12.1. Toxicity

#### **Ecotoxicity effects**

The product contains following substances which are hazardous for the environment.  
Contains a substance which is: Very toxic to aquatic organisms.

| Component | Freshwater Fish                                                                                                                                              | Water Flea          | Freshwater Algae                            |
|-----------|--------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------|---------------------------------------------|
| Nickel    | LC50: > 100 mg/L, 96h<br>(Brachydanio rerio)<br>LC50: = 1.3 mg/L, 96h<br>semi-static (Cyprinus carpio)<br>LC50: = 10.4 mg/L, 96h static<br>(Cyprinus carpio) | EC50 = 510 µg/L 96h | EC50 = 0.1 mg/L 72h<br>EC50 = 0.18 mg/L 72h |

### 12.2. Persistence and degradability

#### **Persistence**

Insoluble in water.

#### **Degradability**

Not relevant for inorganic substances.

#### **Degradation in sewage treatment plant**

Contains substances known to be hazardous to the environment or not degradable in waste water treatment plants.

### 12.3. Bioaccumulative potential

May have some potential to bioaccumulate

### 12.4. Mobility in soil

Spillage unlikely to penetrate soil Is not likely mobile in the environment due its low water solubility.

### 12.5. Results of PBT and vPvB assessment

In accordance with Annex XIII of the REACH Regulation, inorganic substances do not require assessment.

### 12.6. Endocrine disrupting properties

#### **Endocrine Disruptor Information**

This product does not contain any known or suspected endocrine disruptors

### 12.7. Other adverse effects

#### **Persistent Organic Pollutant**

This product does not contain any known or suspected substance

#### **Ozone Depletion Potential**

This product does not contain any known or suspected substance

## SECTION 13: DISPOSAL CONSIDERATIONS

### 13.1. Waste treatment methods

#### **Waste from Residues/Unused Products**

Waste is classified as hazardous. Dispose of in accordance with the European Directives on waste and hazardous waste. Dispose of in accordance with local regulations.

#### **Contaminated Packaging**

Dispose of this container to hazardous or special waste collection point.

#### **European Waste Catalogue (EWC)**

According to the European Waste Catalog, Waste Codes are not product specific, but application specific.

#### **Other Information**

Do not flush to sewer. Waste codes should be assigned by the user based on the application for which the product was used. Do not empty into drains.

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## Switzerland - Waste Ordinance

Disposal should be in accordance with applicable regional, national and local laws and regulations. Ordinance on the Avoidance and the Disposal of Waste (Waste Ordinance, ADWO) SR 814.600  
<https://www.fedlex.admin.ch/eli/cc/2015/891/en>

## SECTION 14: TRANSPORT INFORMATION

### IMDG/IMO

Not regulated

#### 14.1. UN number

#### 14.2. UN proper shipping name

#### 14.3. Transport hazard class(es)

#### 14.4. Packing group

### ADR

Not regulated

#### 14.1. UN number

#### 14.2. UN proper shipping name

#### 14.3. Transport hazard class(es)

#### 14.4. Packing group

### IATA

Not regulated

#### 14.1. UN number

#### 14.2. UN proper shipping name

#### 14.3. Transport hazard class(es)

#### 14.4. Packing group

#### 14.5. Environmental hazards

No hazards identified

#### 14.6. Special precautions for user

No special precautions required.

#### 14.7. Maritime transport in bulk according to IMO instruments

Not applicable, packaged goods

## SECTION 15: REGULATORY INFORMATION

### 15.1. Safety, health and environmental regulations/legislation specific for the substance or mixture

#### International Inventories

Europe (EINECS/ELINCS/NLP), China (IECSC), Taiwan (TCSI), Korea (KECL), Japan (ENCS), Japan (ISHL), Canada (DSL/NDSL), Australia (AICS), New Zealand (NZIoC), Philippines (PICCS). US EPA (TSCA) - Toxic Substances Control Act, (40 CFR Part 710)

| Component | CAS No    | EINECS    | ELINCS | NLP | IECSC | TCSI | KECL     | ENCS | ISHL |
|-----------|-----------|-----------|--------|-----|-------|------|----------|------|------|
| Nickel    | 7440-02-0 | 231-111-4 | -      | -   | X     | X    | KE-25818 | X    | -    |

| Component | CAS No    | TSCA | TSCA Inventory notification - Active-Inactive | DSL | NDSL | AICS | NZIoC | PICCS |
|-----------|-----------|------|-----------------------------------------------|-----|------|------|-------|-------|
| Nickel    | 7440-02-0 | X    | ACTIVE                                        | X   | -    | X    | X     | X     |

Legend: X - Listed '-' - Not Listed

KECL - NIER number or KE number (<http://ncis.nier.go.kr/en/main.do>)

### Authorisation/Restrictions according to EU REACH

| Component | CAS No | REACH (1907/2006) - Annex XIV - Substances Subject to Authorization | REACH (1907/2006) - Annex XVII - Restrictions on Certain Dangerous Substances | REACH Regulation (EC 1907/2006) article 59 - Candidate List of Substances of Very High |
|-----------|--------|---------------------------------------------------------------------|-------------------------------------------------------------------------------|----------------------------------------------------------------------------------------|
|-----------|--------|---------------------------------------------------------------------|-------------------------------------------------------------------------------|----------------------------------------------------------------------------------------|

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|        |           |   |                                                                                                                                          | Concern (SVHC) |
|--------|-----------|---|------------------------------------------------------------------------------------------------------------------------------------------|----------------|
| Nickel | 7440-02-0 | - | Use restricted. See item 27.<br>(see link for restriction details)<br>Use restricted. See item 75.<br>(see link for restriction details) | -              |

## REACH links

<https://echa.europa.eu/substances-restricted-under-reach>

## Seveso III Directive (2012/18/EC)

| Component | CAS No    | Seveso III Directive (2012/18/EC) - Qualifying Quantities for Major Accident Notification | Seveso III Directive (2012/18/EC) - Qualifying Quantities for Safety Report Requirements |
|-----------|-----------|-------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------|
| Nickel    | 7440-02-0 | Not applicable                                                                            | Not applicable                                                                           |

## Regulation (EC) No 649/2012 of the European Parliament and of the Council of 4 July 2012 concerning the export and import of dangerous chemicals

Not applicable

## Contains component(s) that meet a 'definition' of per & poly fluoroalkyl substance (PFAS)?

Not applicable

Take note of Directive 98/24/EC on the protection of the health and safety of workers from the risks related to chemical agents at work .

## National Regulations

**UK** - Take note of Control of Substances Hazardous to Health Regulations (COSHH) 2002 and 2005 Amendment

## WKG Classification

See table for values

| Component | Germany - Water Classification (AwSV) | Germany - TA-Luft Class                                                                                                                   |
|-----------|---------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------|
| Nickel    | WGK 2                                 | Class II : 0.5 mg/m <sup>3</sup> (Massenkonzentration)<br>Krebserzeugende Stoffe - Class II : 0.5 mg/m <sup>3</sup> (Massenkonzentration) |

## Swiss Regulations

Article 4 para. 4 of the Ordinance on the protection of young people in the workplace (SR 822.115) and Article 1 lit. f of the EAER regulation on hazardous work and young people (SR 822.115.2).

Take note on Article 13 Maternity Ordinance (SR 822.111.52) with regards expectant and nursing mothers.

| Component                   | Switzerland - Ordinance on the Reduction of Risk from handling of hazardous substances preparation (SR 814.81) | Switzerland - Ordinance on Incentive Taxes on Volatile Organic Compounds (OVOC) | Switzerland - Ordinance of the Rotterdam Convention on the Prior Informed Consent Procedure |
|-----------------------------|----------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------|
| Nickel<br>7440-02-0 ( >95 ) | Prohibited and Restricted Substances                                                                           |                                                                                 |                                                                                             |

## 15.2. Chemical safety assessment

A Chemical Safety Assessment/Report (CSA/CSR) has not been conducted

## SECTION 16: OTHER INFORMATION

# SAFETY DATA SHEET

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## Full text of H-Statements referred to under sections 2 and 3

H317 - May cause an allergic skin reaction

H351 - Suspected of causing cancer

H372 - Causes damage to organs through prolonged or repeated exposure

## Legend

**CAS** - Chemical Abstracts Service

**EINECS/ELINCS** - European Inventory of Existing Commercial Chemical Substances/EU List of Notified Chemical Substances

**PICCS** - Philippines Inventory of Chemicals and Chemical Substances

**IECSC** - Chinese Inventory of Existing Chemical Substances

**KECL** - Korean Existing and Evaluated Chemical Substances

**TSCA** - United States Toxic Substances Control Act Section 8(b) Inventory

**DSL/NDSL** - Canadian Domestic Substances List/Non-Domestic Substances List

**ENCS** - Japanese Existing and New Chemical Substances

**AICS** - Australian Inventory of Chemical Substances

**NZIoC** - New Zealand Inventory of Chemicals

**WEL** - Workplace Exposure Limit

**ACGIH** - American Conference of Governmental Industrial Hygienists

**DNEL** - Derived No Effect Level

**RPE** - Respiratory Protective Equipment

**LC50** - Lethal Concentration 50%

**NOEC** - No Observed Effect Concentration

**PBT** - Persistent, Bioaccumulative, Toxic

**TWA** - Time Weighted Average

**IARC** - International Agency for Research on Cancer Predicted No Effect Concentration (PNEC)

**LD50** - Lethal Dose 50%

**EC50** - Effective Concentration 50%

**POW** - Partition coefficient Octanol:Water

**vPvB** - very Persistent, very Bioaccumulative

**ADR** - European Agreement Concerning the International Carriage of Dangerous Goods by Road

**IMO/IMDG** - International Maritime Organization/International Maritime Dangerous Goods Code

**OECD** - Organisation for Economic Co-operation and Development

**BCF** - Bioconcentration factor

**ICAO/IATA** - International Civil Aviation Organization/International Air Transport Association

**MARPOL** - International Convention for the Prevention of Pollution from Ships

**ATE** - Acute Toxicity Estimate

**VOC** - (volatile organic compound)

## Key literature references and sources for data

<https://echa.europa.eu/information-on-chemicals>

Suppliers safety data sheet, Chemadvisor - LOLI, Merck index, RTECS

## Training Advice

Chemical hazard awareness training, incorporating labelling, Safety Data Sheets (SDS), Personal Protective Equipment (PPE) and hygiene.

Use of personal protective equipment, covering appropriate selection, compatibility, breakthrough thresholds, care, maintenance, fit and standards.

First aid for chemical exposure, including the use of eye wash and safety showers.

## Prepared By

Health, Safety and Environmental Department

## Creation Date

11-Feb-2010

## Revision Date

10-Feb-2024

## Revision Summary

New emergency telephone response service provider.

**This safety data sheet complies with the requirements of Regulation (EC) No. 1907/2006. COMMISSION REGULATION (EU) 2020/878 amending Annex II to Regulation (EC) No 1907/2006 .**

**For Switzerland - Compiled in accordance with the technical provisions referred to in Annex 2, Number 3, ChemO (SR 813.11 - Ordinance on Protection against Dangerous Substances and Preparations).**

## Disclaimer

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**End of Safety Data Sheet**